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Biological and nutrition data preparation

Week 6: Missingness, outliers, imbalance and scaling, on survey data that has all four.

Dr Haithem Afli

Department of Computer Science

Haithem.afli@mtu.ie

www.mtu.ie

By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO5

- Distinguish missing completely at random, missing at random and missing not at random, and choose a handling strategy accordingly.
- Detect outliers univariately and multivariately, and decide whether to remove, cap or keep them.
- Encode ordinal and nominal categorical variables correctly.
- Select and justify a scaling strategy, and identify which algorithms require it.
- Address class imbalance and explain why resampling belongs inside the cross-validation fold.
- Explain what a survey weight is and why ignoring it changes the conclusion rather than just the precision.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- NHANES combines an interview with a physical examination and laboratory work on a stratified multi-stage sample of the US civilian non-institutionalised population.
- The August 2021 to August 2023 cycle released demographic records for 11,933 participants, of whom 8,860 completed the examination component.
- It is an excellent teaching dataset precisely because it is messy in instructive ways: dietary recall has large measurement error, laboratory values are missing non-randomly, and the survey weights mean that naive row-level analysis gives biased population estimates.

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The method

Part 2 · Concepts, then the mathematics

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Missingness mechanisms

MCAR: missingness unrelated to anything. MAR: explainable by observed variables. MNAR: depends on the unobserved value itself, as when the sickest participants skip the examination. Only MNAR cannot be fixed by imputation alone, and it is the common case in health data.

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Imputation

Mean and median imputation shrink variance and distort correlations. Iterative and k-NN imputation preserve structure better. Always add a missingness indicator column: the fact that a value is absent is often itself predictive.

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Key concepts

Outliers A biological outlier may be an instrument error, a transcription error, or the single most clinically interesting participant in the study. Deleting before understanding is how findings get lost.

Categorical encoding Ordinal variables have a meaningful order and can be integer-coded. Nominal variables must not be, because that invents a distance. One-hot encoding is the default; watch the dimensionality on high-cardinality fields.

Scaling Standardisation subtracts the mean and divides by the standard deviation; min-max maps to a fixed range; robust scaling uses the median and interquartile range and survives outliers.

Class imbalance Random undersampling discards data. Random oversampling duplicates it and encourages overfitting. SMOTE interpolates new minority points, which can be biologically implausible.

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Key concepts (continued)

Measurement variability A 24-hour dietary recall has substantial within-person day-to-day variation. Treating one recall as a person's habitual intake attenuates associations towards the null. This is a property of the instrument, not of the model.

Survey weights NHANES oversamples some subgroups by design. Unweighted analysis describes the sample, not the population. This is a data-preparation decision with scientific consequences.

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Standardisation and the alternatives

$$z = (x - \mu) / \sigma$$
$$x' = (x - min) / (max - min)$$

Standardisation centres and scales by the standard deviation; min-max maps to a fixed range. Robust scaling uses the median and the interquartile range and survives outliers.

μ, σ fitted on the TRAINING fold only, then applied to the validation fold

needed by k-NN, SVM, PCA, and any gradient-based method

not needed by decision trees and forests: the split is threshold-based and univariate

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Missingness mechanisms, and what each permits

In health data MNAR is the common case: the sickest participants are the ones who miss the examination.

Mechanism	Definition	Can imputation fix it?
MCAR	Missingness unrelated to anything	Yes. Complete-case analysis is unbiased but wasteful
MAR	Explainable by observed variables	Yes, if the model uses those variables
MNAR	Depends on the unobserved value itself	No. The information was never recorded

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Writing the pipeline so the order cannot go wrong

If the preprocessing is inside a Pipeline object, the order is correct by construction rather than by discipline.

- sklearn.pipeline.Pipeline: every step is fitted on the training fold only, and applied unchanged to the test fold
- ColumnTransformer: different treatment per column type, so the imputer and the encoder stay inside the same fitted object
- imblearn.pipeline.Pipeline: applies a resampler during fit and skips it during predict, which plain sklearn cannot do
- cross_val_score(pipe, X, y, cv=StratifiedKFold(...)): refits the whole pipeline on each fold, which is the point
- The test set is never an argument to fit(), fit_transform() or fit_resample(). If it is, something has leaked

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Imbalance, and survey weights

Measurement error in the exposure is usually the binding constraint in nutrition research, not model capacity.

- Random undersampling discards data; random oversampling duplicates it and encourages overfitting
- SMOTE interpolates new minority points, which can be biologically implausible
- All resampling belongs inside the fold, never before the split
- Report balanced accuracy and per-class recall, not accuracy
- NHANES oversamples some subgroups by design
- Unweighted analysis describes the sample, not the population: a data-preparation decision with scientific consequences

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Worked example

Part 3 · The order of operations, and why it is not arbitrary

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The order of operations, and why it is not arbitrary

- A plausible-looking pipeline: impute missing values, remove outliers, scale, balance the classes with SMOTE, split into train and test, cross-validate.
- Every step before the split has leaked. The imputer saw test rows. The outlier rule was set using test rows. The scaler's mean and variance include test rows.
- Worst of all, SMOTE generated synthetic minority points by interpolating between real points, and those synthetic points can land in the test fold, interpolated from training points. The reported score can be dramatically optimistic.
- The correct order: split first, then build a Pipeline containing the imputer, the encoder, the scaler and the resampler, and let cross-validation refit that whole pipeline on each training fold.

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The order of operations, and why it is not arbitrary (continued)

- In scikit-learn this means using imblearn.pipeline.Pipeline rather than sklearn.pipeline.Pipeline, because the former applies the resampler only during fit and not during predict, which is exactly what you want.

- Outlier removal deserves a separate remark. Removing outliers from the test set is not leakage, it is worse: it is quietly deleting the hard cases and then reporting how well you did on the easy ones.

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Split first. Everything that learns anything from data goes inside the pipeline.

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Try this interactively

Data Quality Explorer

What does each preprocessing choice cost, when the damage is known?

`weeks/week06.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · What pretraining does not fix

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What pretraining does not fix

- It is tempting to assume a large pretrained model absorbs messy data gracefully. For the parts of messiness that are about representation, such as unusual phrasing in clinical notes, that is partly true. For the parts that are about the data-generating process, it is not true at all.

- If your outcome is missing not at random, no amount of pretraining recovers the missing information, because the information was never collected. If your cohort under-represents a population, a pretrained model will not tell you so; it will simply be less accurate for that group, silently.

- Data quality is upstream of model choice.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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You find that serum vitamin D is missing for 18 per cent of NHANES participants, and that missingness is more common among participants who did not attend the examination centre. What kind of missingness is this most likely to be, and what does it do to a model trained only on complete cases?

Think like a scientist. Bring your answer to the laboratory.

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Preparing NHANES without breaking it

This week's laboratory · Take real survey data from raw files to a modelling-ready matrix, making every preprocessing decision explicitly. · 2 hours

- Download the demographic file and one laboratory or dietary file for the August 2021 to August 2023 cycle. Merge on SEQN. Report how many participants survive the merge and why the number fell.
- Profile the missingness. For your chosen outcome, test whether missingness is associated with age, sex or examination attendance. Classify the mechanism as MCAR, MAR or MNAR and justify it.
- Handle the missingness three ways: complete-case, median imputation, and iterative imputation with a missingness indicator. Keep all three datasets.
- Identify outliers univariately with the IQR rule and multivariately with IsolationForest. For three of the flagged participants, decide whether to keep them and write one sentence of justification each.
- Encode the categorical variables correctly, distinguishing ordinal from nominal. Scale appropriately for a downstream logistic regression.
- Fit the same model to all three imputation variants inside a proper Pipeline. Report the three cross-validated scores and say which preprocessing choice you would defend in a paper, and why.
- Finally, compute a weighted and an unweighted prevalence estimate for your outcome using WTMEC2YR. Explain the difference in one paragraph.

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Self-check

- Where in the workflow should SMOTE be applied?
- A) On the full dataset before splitting
- B) Inside the cross-validation fold, on the training portion only
- C) On the test set only
- D) After model training

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Self-check

- Which scaler is most appropriate when a feature has a heavy right tail and several extreme values you want to keep?
- A) StandardScaler
- B) MinMaxScaler
- C) RobustScaler
- D) No scaling

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References and further reading

Centers for Disease Control and Prevention, National Center for Health Statistics. National Health and Nutrition Examination Survey, August 2021 - August 2023.

Kapoor, S. & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4(9):100804. doi:10.1016/j.patter.2023.100804

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Questions

Next week: continue to Week 7.

Dr Haithem Afli · haithem.afli@mtu.ie

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